SOFTWARE REQUIREMENT SPECIFICATION

I N V E R N O V A

University Internship & Industry Matching Portal

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University Internship & Industry Matching Portal

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the University Internship & Industry Matching Portal. This document serves as a reference for developers, testers, project supervisors, and stakeholders during system development and validation.

1.2 Scope

The system is a web-based platform connecting Students, Companies, and University Administrators. It provides two main modules:

1. Internship Management
2. Projects & Competitions Management

1.3 Definitions

- **SRS** - Software Requirements Specification
- **UI** - User Interface
- **API** - Application Programming Interface
- **RBAC** - Role-Based Access Control

2. Overall Description

2.1 Product Perspective

- **Frontend:** React
- **Backend:** .NET10
- **Database:** MySQL
- **Deployment:** Microsoft Azure
- **Architecture:** REST API

2.2 User Classes

- **Students:** Apply for internships, join projects, participate in competitions.
- **Companies:** Submit internships, review applicants, schedule interviews.
- **Administrators:** Approve companies, internships, projects, and generate reports.

3. Functional Requirements

3.1 User Authentication

- **FR-AUTH-01:** The system shall allow users to register.
- **FR-AUTH-02:** The system shall authenticate users securely.
- **FR-AUTH-03:** The system shall implement role-based access control.
- **FR-AUTH-04:** The system shall allow password reset.

3.2 Internship Management

- **FR-INT-01:** Companies shall submit internship postings.
- **FR-INT-02:** Admin shall approve internships before publication.
- **FR-INT-03:** Only approved internships shall be visible to students.
- **FR-INT-04:** Students shall apply for internships.
- **FR-INT-05:** The system shall track application status (Applied → Shortlisted → Interview Scheduled → Selected/Rejected).

3.3 Interview Scheduling

- **FR-SCH-01:** Companies shall schedule interviews.
- **FR-SCH-02:** Students shall receive interview notifications.

3.4 Resume Management

- **FR-DOC-01:** Students shall upload resumes and documents.
- **FR-DOC-02:** Companies shall securely download applicant documents.

3.5 Project Module

- **FR-PROJ-01:** Admin shall approve project postings.
- **FR-PROJ-02:** Students shall apply to join projects.
- **FR-PROJ-03:** Project leaders shall accept or reject participants.

3.6 Competition Module

- **FR-COMP-01:** Admin shall approve competition postings.
- **FR-COMP-02:** Students shall register for competitions.

3.7 Notifications

- **FR-NOT-01:** The system shall notify users when application status changes.

3.8 Logging

- **FR-LOG-01:** The system shall log login attempts and approval actions.

4. Data Requirements

Entities:

- **Student:** (StudentID, Name, Email, Skills, ResumePath)
- **Company:** (CompanyID, Name, Email, VerificationStatus)
- **Internship:** (InternshipID, Title, Description, RequiredSkills, Status)
- **Application:** (ApplicationID, StudentID, InternshipID, Status)
- **Project:** (ProjectID, Title, LeaderID)
- **Competition:** (CompetitionID, Title, Organizer, Deadline)

5. Non-Functional Requirements

Performance

- **NFR-PERF-01:** System response time shall be under 3 seconds.

- **NFR-PERF-02:** System shall support at least 500 concurrent users.

Security

- **NFR-SEC-01:** Passwords shall be encrypted.
- **NFR-SEC-02:** All communication shall use HTTPS.

Reliability

- **NFR-REL-01:** System uptime shall be 99% or higher.

Usability

- **NFR-USA-01:** The system shall provide a mobile-responsive UI.

6. Verification and Validation

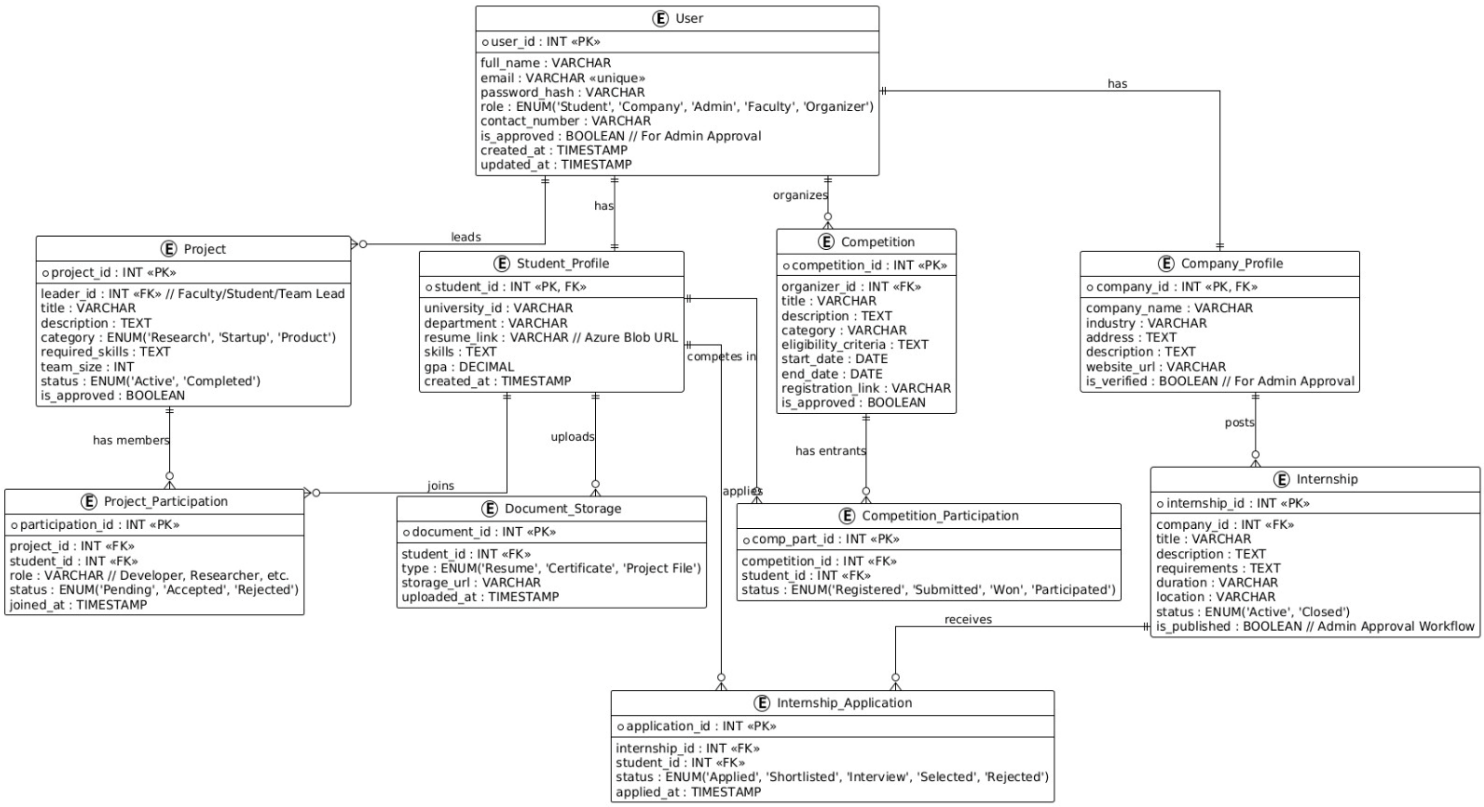
The system will undergo Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).

7. Acceptance Criteria

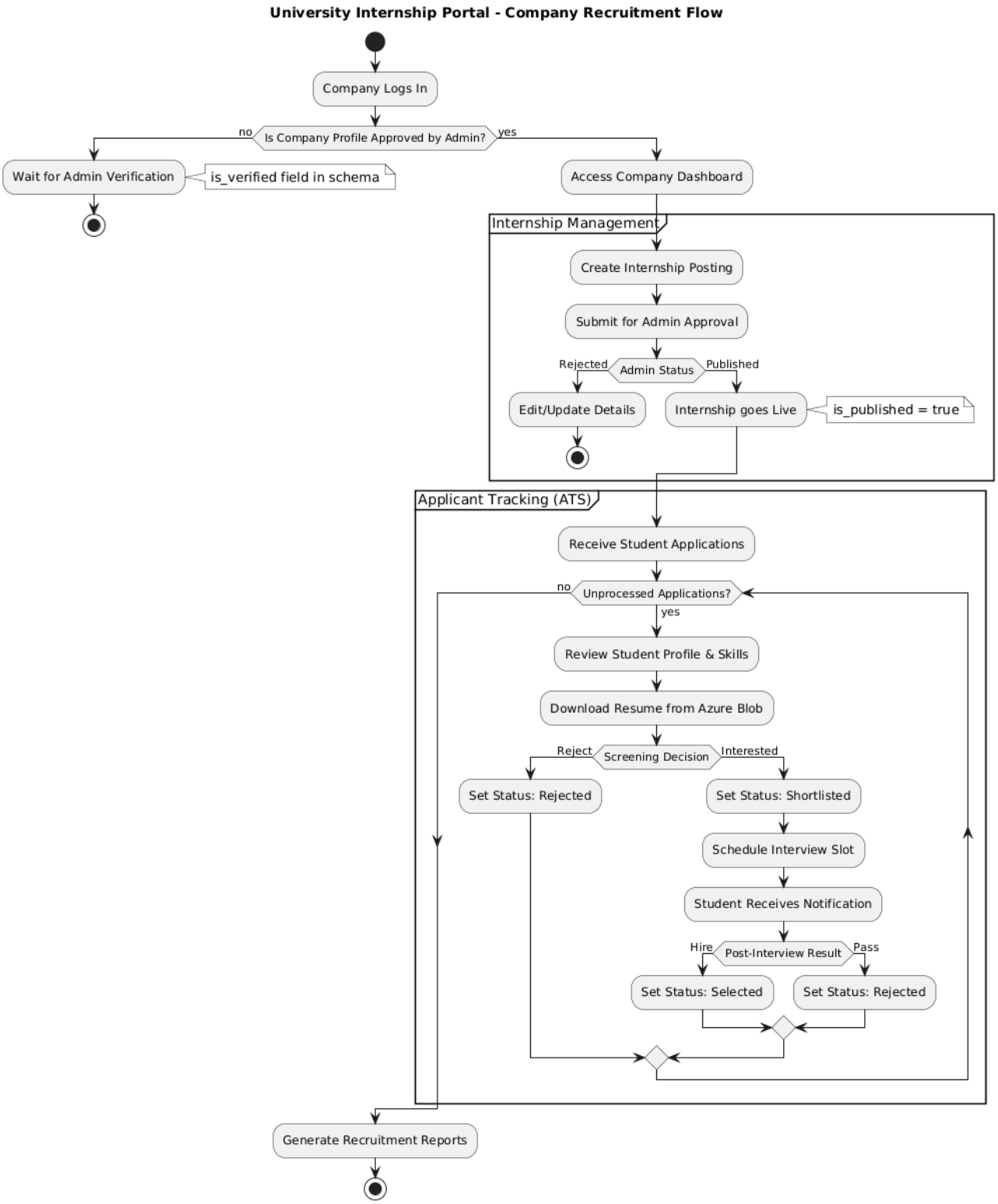
The system will be accepted when:

- All functional requirements are implemented.
- Internship workflow functions end-to-end.
- Deployment on Azure is successful.
- No critical defects remain.

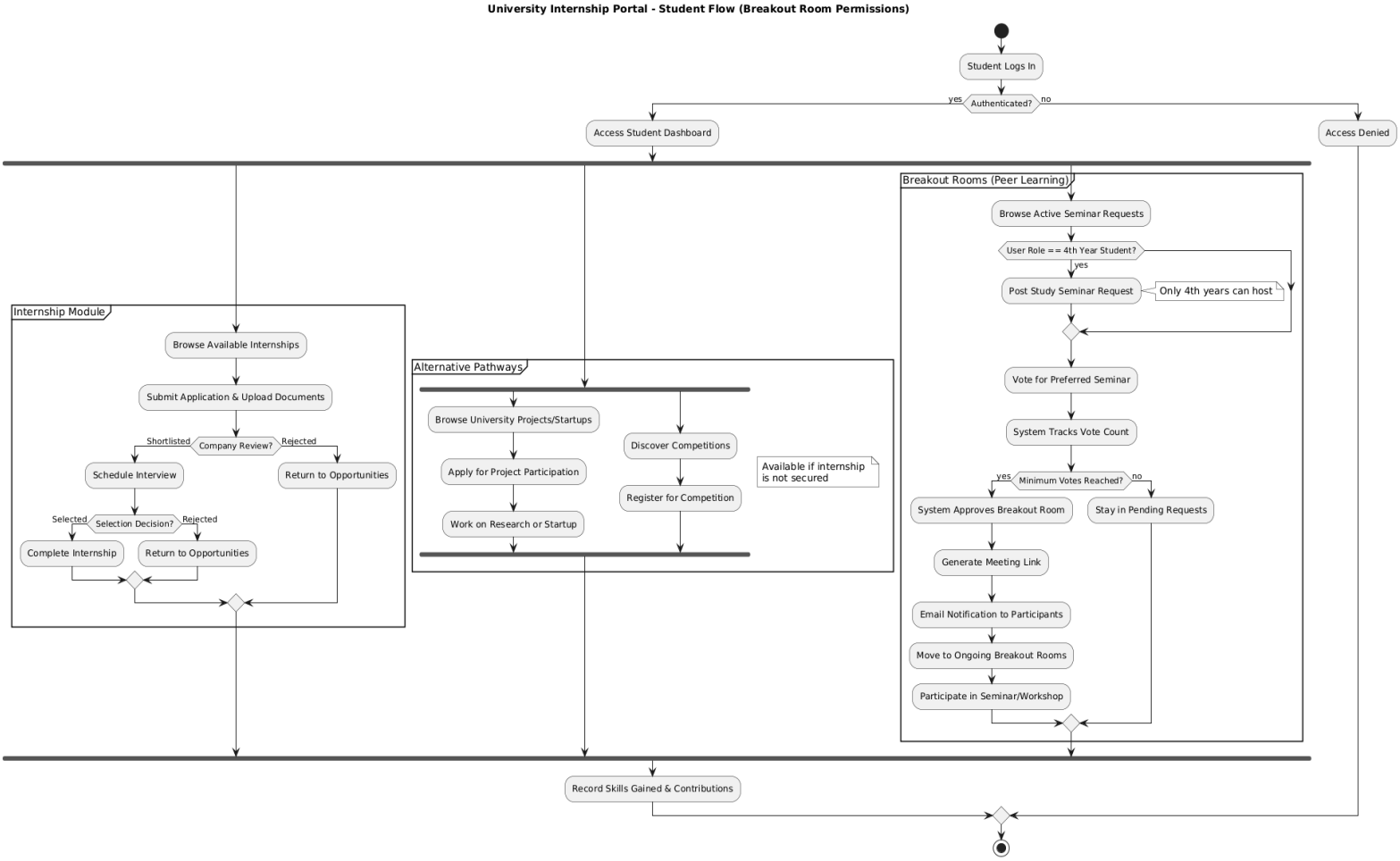
8. Class diagram



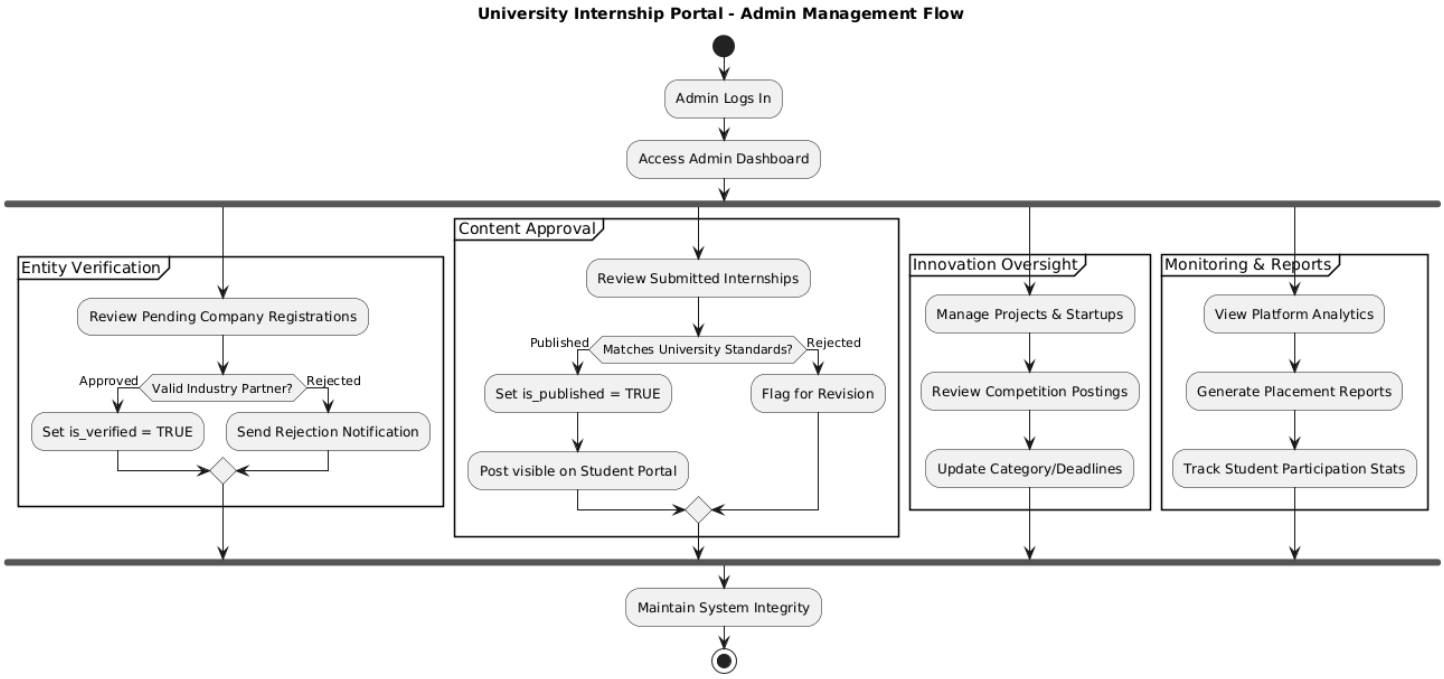
8. Company Control flow diagram



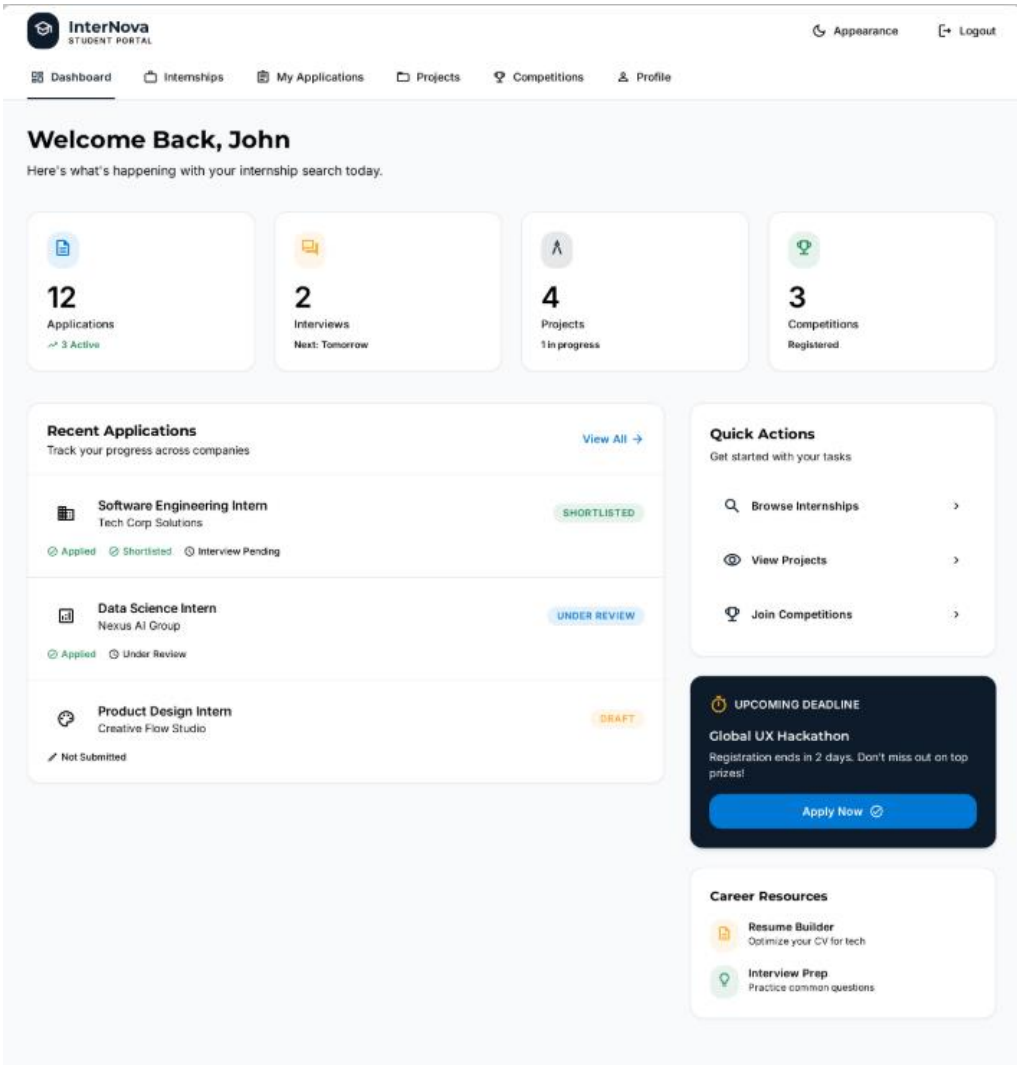
9. Student control flow diagram

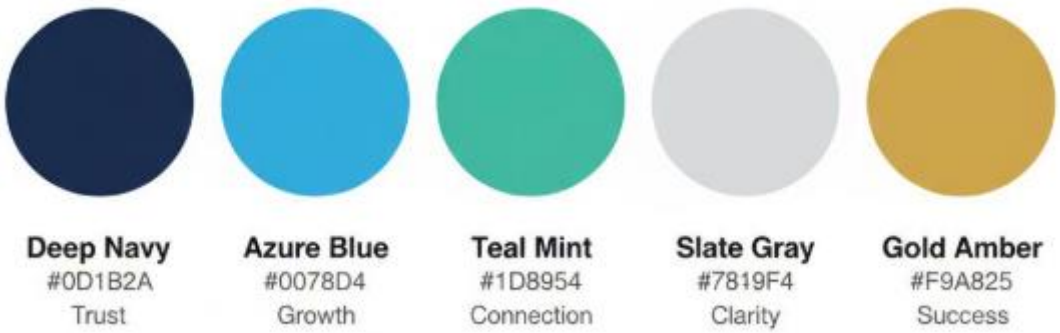


10. Admin control flow diagram



11. Color palate and sample UI design





12. Project Assumptions

- It is assumed that all user classes (Students, Companies, and Administrators) have access to a stable internet connection to use the web-based platform.
- There is an assumption that the **Administrator** will manually verify and approve companies and internship postings before they become visible to students.
- The system assumes that students will upload resumes in a format that companies can successfully download and view.
- It is assumed the hosting environment will maintain a system uptime of 99% or higher.